

Likelihood Assignment 2

```
library(readr)
```

Warning: package 'readr' was built under R version 4.4.1

```
coffeetime <- read_csv("coffeetime.csv")
```

Rows: 98 Columns: 2

-- Column specification -----

Delimiter: ","

dbl (2): time, cafe

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
View(coffeetime)
```

```
library(cmdstanr)
```

This is cmdstanr version 0.8.0

- CmdStanR documentation and vignettes: mc-stan.org/cmdstanr

- CmdStan path: C:/Users/emmaw/.cmdstan/cmdstan-2.38.0

- CmdStan version: 2.38.0

A newer version of CmdStan is available. See `?install_cmdstan()` to install it.

To disable this check set option or environment variable `cmdstanr_no_ver_check=TRUE`.

Question 1

By conditional independence of the T_i 's given λ ,

$$\pi(t \mid \lambda) = \prod_{i=1}^n \pi(t_i \mid \lambda_i).$$

Also, since the λ_i 's are iid given $\theta = (\alpha, \beta)$,

$$\pi(\lambda \mid \theta) = \prod_{i=1}^n \pi(\lambda_i \mid \theta).$$

Therefore,

$$\pi(t \mid \lambda) \pi(\lambda \mid \theta) = \prod_{i=1}^n \pi(t_i \mid \lambda_i) \pi(\lambda_i \mid \theta).$$

Each term in the product depends only on one λ_i , so the integral factorises:

$$\begin{aligned} L(\theta) &= \int \prod_{i=1}^n \pi(t_i \mid \lambda_i) \pi(\lambda_i \mid \theta) d\lambda_1 \cdots d\lambda_n \\ &= \prod_{i=1}^n \int_0^\infty \pi(t_i \mid \lambda_i) \pi(\lambda_i \mid \theta) d\lambda_i. \end{aligned}$$

Question 3

For repeated visits to café (i), the model is:

$$T_{ij} \mid \lambda_i \sim \text{Exponential}(\lambda_i), \quad j = 1, \dots, v_i, \quad i = 1, \dots, n.$$

The café-specific expected waiting times are modelled as

$$\lambda_i \stackrel{iid}{\sim} \text{Gamma}(\alpha, \beta), \quad i = 1, \dots, n.$$

Conditional on λ_i , the waiting times from repeated visits to the same café are assumed independent.

Question 4

Let $t_i = (t_{i1}, \dots, t_{iv_i})$ denote the waiting times observed at café (i).

The likelihood is obtained by marginalising over the unobserved λ_i 's:

$$L(\theta) = \pi(t \mid \theta) = \int \pi(t \mid \lambda) \pi(\lambda \mid \theta) d\lambda.$$

Given λ , observations from different cafés are independent, so

$$\pi(t \mid \lambda) = \prod_{i=1}^n \pi(t_i \mid \lambda_i).$$

Within café (i), repeated visits are independent conditional on λ_i , hence

$$\pi(t_i \mid \lambda_i) = \prod_{j=1}^{v_i} \pi(t_{ij} \mid \lambda_i).$$

As before, the λ_i 's are iid, therefore:

$$\pi(\lambda \mid \theta) = \prod_{i=1}^n \pi(\lambda_i \mid \theta).$$

Therefore,

$$L(\theta) = \int \prod_{i=1}^n \left[\left\{ \prod_{j=1}^{v_i} \pi(t_{ij} \mid \lambda_i) \right\} \pi(\lambda_i \mid \theta) \right] d\lambda_1 \cdots d\lambda_n.$$

Since each bracketed term depends only on one λ_i , the integral separates:

$$L(\theta) = \prod_{i=1}^n \int_0^\infty \left[\prod_{j=1}^{v_i} \pi(t_{ij} \mid \lambda_i) \right] \pi(\lambda_i \mid \theta) d\lambda_i.$$

Q5

```
mu <- mean(coffeetime$time)
sigma <- sd(coffeetime$time)

alpha_prior <- mu^{2}/ sigma^{2}
beta_pror <- mu/ sigma^2
```

It is expected that the average café waiting time will be around 4.4 minutes, with substantial variation between cafés, perhaps corresponding to a standard deviation of roughly 5 minutes. Matching these moments to a Gamma distribution gives approximate hyperparameter values of $\alpha \approx 0.78$ and $\beta \approx 0.18$.

I therefore place exponential priors on the Gamma hyperparameters:

$$\alpha \sim \text{Exponential}\left(\frac{1}{0.78}\right), \quad \beta \sim \text{Exponential}\left(\frac{1}{0.18}\right).$$

These priors are relatively weakly informative, while still centering the model around plausible waiting times and between-café variability.

```
random_effects_stan <- "
data {
  int<lower=0> N;
  vector<lower=0>[N] t;
  int<lower=1> n_cafes;
  array[N] int<lower=1, upper=n_cafes> cafe;
}

parameters {
  real<lower=0> alpha;
  real<lower=0> beta;
  vector<lower=0>[n_cafes] lambda;
}

model {
  alpha ~ exponential(1 / 0.7824712);
  beta ~ exponential(1 / 0.1764025);

  lambda ~ gamma(alpha, beta);

  for (k in 1:N) {
    t[k] ~ exponential(1 / lambda[cafe[k]]);
  }
}
```

```

}
"

writeLines(random_effects_stan, "random_effects_model.stan")

model <- cmdstan_model("random_effects_model.stan")

```

In file included from stan/src/stan/model/model_header.hpp:5:

stan/src/stan/model/model_base_crtp.hpp:136:8: warning: 'void stan::model::model_base_crtp<M, 1, 1>; std::ostream = std::basic_ostream<char>]' was hidden [-Woverloaded-virtual=]

```

136 | void write_array(stan::rng_t& rng, Eigen::VectorXd& theta,
    | ~~~~~

```

C:/Users/emmaw/AppData/Local/Temp/RtmpigQ7Kl/model-d4ac21b15497.hpp:442:3: note: by 'void stan::model::model_base_crtp<M, 1, 1>&, Eigen::Matrix<double, -1, 1>&, bool, bool, std::ostream*) const'

```

442 | write_array(RNG& base_rng, Eigen::Matrix<double, -1, 1>& params_r,
    | ~~~~~

```

stan/src/stan/model/model_base_crtp.hpp:205:8: warning: 'void stan::model::model_base_crtp<M, 1, 1>; std::vector<double>& theta, std::vector<int>& theta_i]' was hidden [-Woverloaded-virtual=]

```

205 | void write_array(stan::rng_t& rng, std::vector<double>& theta,
    | ~~~~~

```

C:/Users/emmaw/AppData/Local/Temp/RtmpigQ7Kl/model-d4ac21b15497.hpp:442:3: note: by 'void stan::model::model_base_crtp<M, 1, 1>&, Eigen::Matrix<double, -1, 1>&, bool, bool, std::ostream*) const'

```

442 | write_array(RNG& base_rng, Eigen::Matrix<double, -1, 1>& params_r,
    | ~~~~~

```

stan/src/stan/model/model_base_crtp.hpp:154:17: warning: 'double stan::model::model_base_crtp<M, 1, 1>; std::vector<double>& theta, std::vector<int>& theta_i]' was hidden [-Woverloaded-virtual=]

```

154 | inline double log_prob(std::vector<double>& theta, std::vector<int>& theta_i,
    | ~~~~~

```

C:/Users/emmaw/AppData/Local/Temp/RtmpigQ7Kl/model-d4ac21b15497.hpp:475:3: note: by 'T_ ran 1, 1>&, std::ostream*) const'

```

475 | log_prob(Eigen::Matrix<T_, -1, 1>& params_r, std::ostream* pstream = nu

```

```

llptr) const {
    | ~~~~~

```

stan/src/stan/model/model_base_crtp.hpp:91:17: warning: 'double stan::model::model_base_crtp<M, 1, 1>; std::vector<double>& theta, std::vector<int>& theta_i]' was hidden [-Woverloaded-virtual=]

```

91 | inline double log_prob(Eigen::VectorXd& theta,
    | ~~~~~

```

```

C:/Users/emmaw/AppData/Local/Temp/RtmpigQ7Kl/model-d4ac21b15497.hpp:475:3: note:   by 'T_ ran
1, 1>&, std::ostream*) const'
    475 |     log_prob(Eigen::Matrix<T_,-1,1>& params_r, std::ostream* pstream = nullptr) const
        |     ^~~~~~

stan/src/stan/model/model_base_crtp.hpp:159:20: warning: 'stan::math::var stan::model::model_l
Woverloaded-virtual=]
    159 |     inline math::var log_prob(std::vector<math::var>& theta,
        |                               ^~~~~~

C:/Users/emmaw/AppData/Local/Temp/RtmpigQ7Kl/model-d4ac21b15497.hpp:475:3: note:   by 'T_ ran
1, 1>&, std::ostream*) const'
    475 |     log_prob(Eigen::Matrix<T_,-1,1>& params_r, std::ostream* pstream = nullptr) const
        |     ^~~~~~

stan/src/stan/model/model_base_crtp.hpp:96:20: warning: 'stan::math::var stan::model::model_l
1, 1>&, std::ostream*) const [with M = random_effects_model_model_namespace::random_effects_r
Woverloaded-virtual=]
     96 |     inline math::var log_prob(Eigen::Matrix<math::var, -1, 1>& theta,
        |                               ^~~~~~

C:/Users/emmaw/AppData/Local/Temp/RtmpigQ7Kl/model-d4ac21b15497.hpp:475:3: note:   by 'T_ ran
1, 1>&, std::ostream*) const'
    475 |     log_prob(Eigen::Matrix<T_,-1,1>& params_r, std::ostream* pstream = nullptr) const
        |     ^~~~~~

```

```

stan_data <- list(
  N = length(coffeetime$cafe),
  n_cafes = 15,
  t = coffeetime$time,
  cafe = coffeetime$cafe
)

```

```

fit <- model$sample(
  data = stan_data,
  chains = 4,
  iter_warmup = 1000,
  iter_sampling = 1000,
  seed = 123
)

```

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)

```
Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
```

Chain 1 Informational Message: The current Metropolis proposal is about to be rejected because of the following reasons:

Chain 1 Exception: gamma_lpdf: Random variable[2] is inf, but must be positive finite! (in 'd4ac21b15497.stan', line 19, column 2 to column 30)

Chain 1 If this warning occurs sporadically, such as for highly constrained variable types like integer, the error message can be safely ignored.

Chain 1 but if this warning occurs often then your model may be either severely ill-conditioned or misspecified.

Chain 1

```
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.3 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
```

Chain 2 Informational Message: The current Metropolis proposal is about to be rejected because

Chain 2 Exception: gamma_lpdf: Random variable[4] is inf, but must be positive finite! (in 'd4ac21b15497.stan', line 19, column 2 to column 30)

Chain 2 If this warning occurs sporadically, such as for highly constrained variable types like

Chain 2 but if this warning occurs often then your model may be either severely ill-conditioned or misspecified.

Chain 2

Chain 2 Iteration: 600 / 2000 [30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.3 seconds.

Chain 3 Iteration: 1 / 2000 [0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [35%] (Warmup)

Chain 3 Informational Message: The current Metropolis proposal is about to be rejected because

Chain 3 Exception: gamma_lpdf: Random variable[1] is inf, but must be positive finite! (in 'C:\Users\user\AppData\Local\Temp\d4ac21b15497.stan', line 19, column 2 to column 30)

Chain 3 If this warning occurs sporadically, such as for highly constrained variable types 1.

Chain 3 but if this warning occurs often then your model may be either severely ill-conditioned or misspecified.

Chain 3

Chain 3 Iteration: 800 / 2000 [40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.2 seconds.
Chain 4 Iteration: 1 / 2000 [0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)

Chain 4 Informational Message: The current Metropolis proposal is about to be rejected because of the following reasons:

Chain 4 Exception: gamma_lpdf: Random variable[1] is inf, but must be positive finite! (in 'C:\Users\user\AppData\Local\Temp\d4ac21b15497.stan', line 19, column 2 to column 30)

Chain 4 If this warning occurs sporadically, such as for highly constrained variable types 1.

Chain 4 but if this warning occurs often then your model may be either severely ill-conditioned or misspecified.

Chain 4

```
Chain 4 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.3 seconds.
```

All 4 chains finished successfully.

Mean chain execution time: 0.3 seconds.

Total execution time: 1.6 seconds.

```
sum <- fit$summary()

sum[] <- lapply(sum, function(x) {
  if(is.numeric(x)) round(x, 3) else x
})

kableExtra::kable(sum)
```

variable	mean	median	sd	mad	q5	q95	rhat	ess_bulk	ess_tail
lp__	-	-	2.957	2.902	-	-	1.001	1691.433	2349.490
	256.640	256.364			261.937	252.350			
alpha	2.049	1.951	0.709	0.669	1.064	3.400	1.001	2954.681	2947.611
beta	0.400	0.378	0.160	0.152	0.176	0.698	1.001	2952.544	2916.952
lambda[1]	5.190	4.944	1.597	1.447	3.118	8.069	1.002	7248.406	2727.542
lambda[2]	6.266	5.548	3.205	2.555	2.612	12.195	1.000	8654.841	2444.857

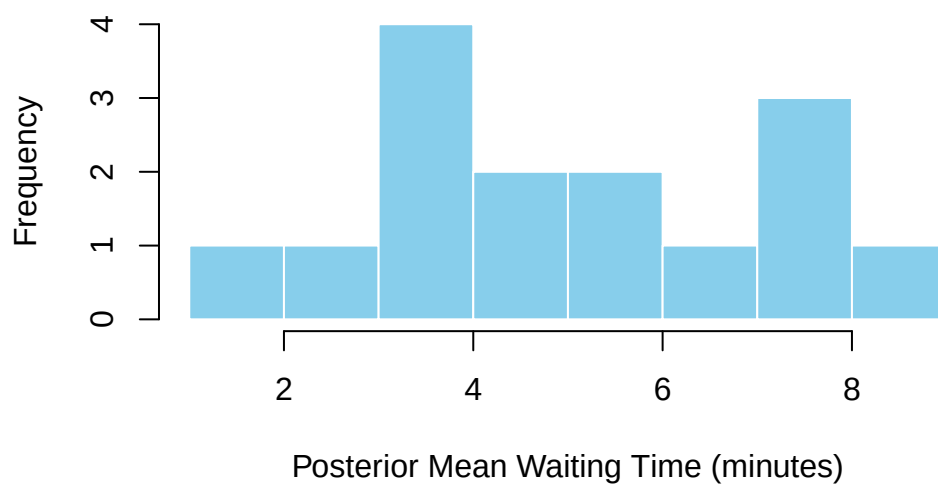
variable	mean	median	sd	mad	q5	q95	rhat	ess_bulk	ess_tail
lambda[3]	4.020	3.732	1.484	1.243	2.213	6.692	1.001	7811.450	2855.724
lambda[4]	7.264	6.638	2.945	2.550	3.688	12.836	1.001	7494.436	3131.874
lambda[5]	5.321	4.963	1.986	1.713	2.893	8.916	1.000	8382.607	2220.773
lambda[6]	7.972	7.510	2.615	2.309	4.596	12.907	1.001	6683.525	3033.366
lambda[7]	2.719	2.409	1.362	1.038	1.193	5.305	1.002	7067.826	3008.157
lambda[8]	1.459	1.246	0.844	0.632	0.580	3.119	1.001	6868.141	2884.967
lambda[9]	4.981	4.704	1.615	1.399	2.931	7.993	1.004	6744.271	2809.441
lambda[10]	8.857	8.215	3.433	2.900	4.630	15.296	1.000	6148.502	3130.779
lambda[11]	3.719	3.425	1.453	1.234	1.958	6.503	1.003	8001.725	2359.246
lambda[12]	3.445	3.272	1.141	1.027	1.974	5.600	1.001	8641.370	2169.160
lambda[13]	3.754	3.519	1.275	1.082	2.174	6.182	1.003	7893.678	3243.142
lambda[14]	7.010	6.567	2.511	2.146	3.876	11.641	1.001	7386.147	2939.965
lambda[15]	3.004	2.677	1.497	1.205	1.330	5.817	1.000	7469.603	2970.038

The MCMC diagnostics suggest that the random effects model has fitted successfully. The $\hat{R}$ values are close to 1, which suggests that the chains have mixed well and are sampling from the same posterior distribution. The effective sample sizes are also reasonably large, meaning that the posterior summaries should be reliable. There is therefore no obvious evidence of non-convergence or poor mixing.

The posterior distribution of the café-specific expected waiting times shows that there is variation between cafés. This supports the use of a random effects model, since the model allows each café to have its own expected waiting time while still assuming that the cafés come from a common population distribution.

```
hist(
  sum$mean[4:18],
  main = "Posterior Mean Waiting Times by Café",
  xlab = "Posterior Mean Waiting Time (minutes)",
  ylab = "Frequency",
  border = "white",
  col = "skyblue",
  breaks = 8
)
```

Posterior Mean Waiting Times by Café



```
library(posterior)
```

Warning: package 'posterior' was built under R version 4.4.3

This is posterior version 1.6.1

Attaching package: 'posterior'

The following objects are masked from 'package:stats':

mad, sd, var

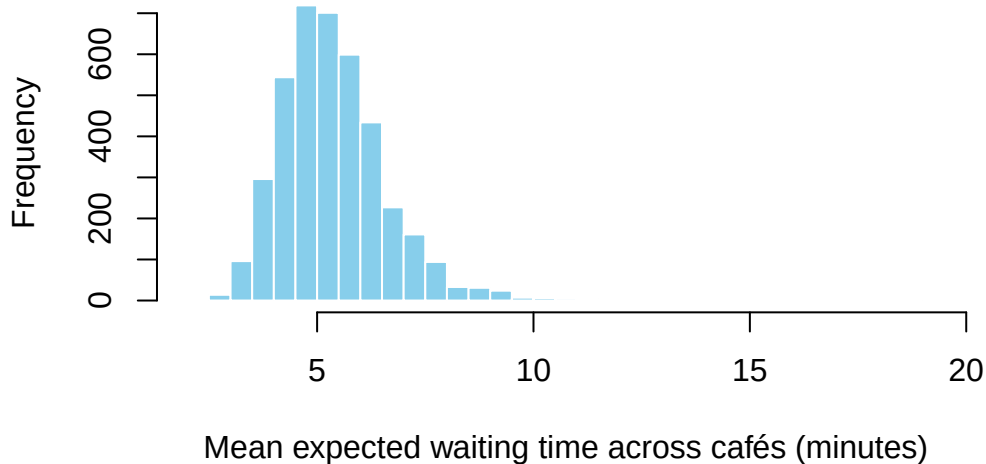
The following objects are masked from 'package:base':

%in%, match

```
samples <- fit$draws(c("alpha", "beta"), format = "matrix")  
mean_waiting_time <- samples[, "alpha"] / samples[, "beta"]
```

```
hist(
  mean_waiting_time,
  main = "Posterior Draws of the Mean Expected Waiting Time Across Cafés",
  xlab = "Mean expected waiting time across cafés (minutes)",
  ylab = "Frequency",
  breaks = 30,
  col = "skyblue",
  border = "white"
)
```

Posterior Draws of the Mean Expected Waiting Time Across Cafés



Q7

For the fixed effects model, we still allow the expected waiting time to differ by café, but we do not assume that the café-specific parameters are drawn from a shared Gamma distribution. Thus, there is no pooling of information between cafés.

Let T_{ij} be the waiting time for visit j to café i . I specify $T_{ij} \mid \lambda_i \sim \text{Exponential}(\lambda_i)$, $j = 1, \dots, v_i$, $i = 1, \dots, n$

so that $(T_{ij} \mid \lambda_i) = \lambda_i$.

For the café-specific expected waiting times, independent priors are used:

$$\lambda_i \overset{ind}{\sim} \text{Exponential}(4.4) \quad i = 1, \dots, n$$

I chose 4.4 because in the random effects model, I expected the average café waiting time to be around 4.4 minutes. However, unlike the random effects model, this fixed effects model does not estimate shared parameters such as α and β . Therefore the cafés are not linked through a common estimated population distribution and the posterior estimate of each λ_i is informed only by the data from only café i and its own prior.

Q8

```
fixed_effects_stan <- "
data {
  int<lower=0> N;
  vector<lower=0>[N] t;
  int<lower=1> n_cafes;
  array[N] int<lower=1, upper=n_cafes> cafe;
}

parameters {
  vector<lower=0>[n_cafes] lambda;
}

model {
  lambda ~ exponential(1 / 4.4);

  for (k in 1:N) {
    t[k] ~ exponential(1 / lambda[cafe[k]]);
  }
}
"

writeLines(fixed_effects_stan, "fixed_effects_model.stan")

fixed_mod <- cmdstan_model("fixed_effects_model.stan")
```

In file included from stan/src/stan/model/model_header.hpp:5:

stan/src/stan/model/model_base_crtp.hpp:136:8: warning: 'void stan::model::model_base_crtp<M, 1, 1>; std::ostream = std::basic_ostream<char>]' was hidden [-Woverloaded-
virtual=]

```
136 | void write_array(stan::rng_t& rng, Eigen::VectorXd& theta,
    | ~~~~~
```



```

    96 | inline math::var log_prob(Eigen::Matrix<math::var, -1, 1>& theta,
      |           ^~~~~~
C:/Users/emmaw/AppData/Local/Temp/RtmpigQ7Kl/model-d4ac543a5049.hpp:409:3: note: by 'T_ fi
1, 1>&, std::ostream*) const'
    409 | log_prob(Eigen::Matrix<T_,-1,1>& params_r, std::ostream* pstream = nullptr) const
      |           ^~~~~~

```

```
library(tidyverse)
```

Warning: package 'tidyverse' was built under R version 4.4.3

Warning: package 'ggplot2' was built under R version 4.4.3

Warning: package 'tidyr' was built under R version 4.4.3

Warning: package 'dplyr' was built under R version 4.4.3

Warning: package 'stringr' was built under R version 4.4.3

Warning: package 'forcats' was built under R version 4.4.3

Warning: package 'lubridate' was built under R version 4.4.3

```

-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v purrr      1.0.2
v forcats    1.0.0      v stringr    1.5.1
v ggplot2    4.0.2      v tibble     3.2.1
v lubridate  1.9.5      v tidyr      1.3.1
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become

```

```

# Fit model
fixed_fit <- fixed_mod$sample(
  data = stan_data,
  chains = 4,
  iter_warmup = 1000,
  iter_sampling = 1000,
  seed = 123
)

```

Running MCMC with 4 sequential chains...

```
Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration:   700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration:   900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
```

```
Chain 1 Informational Message: The current Metropolis proposal is about to be rejected because of the following reasons:
Chain 1 Exception: exponential_lpdf: Inverse scale parameter is 0, but must be positive finite (in 'd4ac543a5049.stan', line 17, column 4 to column 44)
Chain 1 If this warning occurs sporadically, such as for highly constrained variable types like real, it is mostly harmless
Chain 1 but if this warning occurs often then your model may be either severely ill-conditioned or misspecified.
Chain 1
```

```
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.1 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
```

```
Chain 2 Informational Message: The current Metropolis proposal is about to be rejected because of the following reasons:
Chain 2 Exception: exponential_lpdf: Inverse scale parameter is 0, but must be positive finite (in 'd4ac543a5049.stan', line 17, column 4 to column 44)
Chain 2 If this warning occurs sporadically, such as for highly constrained variable types like real, it is mostly harmless
Chain 2 but if this warning occurs often then your model may be either severely ill-conditioned or misspecified.
```

Chain 2

```
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.2 seconds.
Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
```

Chain 3 Informational Message: The current Metropolis proposal is about to be rejected because of the following reasons:

Chain 3 Exception: exponential_lpdf: Inverse scale parameter is 0, but must be positive finite (in 'd4ac543a5049.stan', line 17, column 4 to column 44)

Chain 3 If this warning occurs sporadically, such as for highly constrained variable types 1.
Chain 3 but if this warning occurs often then your model may be either severely ill-
conditioned or misspecified.

Chain 3

Chain 3 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)

Chain 3 finished in 0.1 seconds.

Chain 4 Iteration: 1 / 2000 [0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [20%] (Warmup)
Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)

Chain 4 Informational Message: The current Metropolis proposal is about to be rejected because of the following reasons:
Chain 4 Exception: exponential_lpdf: Inverse scale parameter is 0, but must be positive finite (in 'd4ac543a5049.stan', line 17, column 4 to column 44)

Chain 4 If this warning occurs sporadically, such as for highly constrained variable types 1.
Chain 4 but if this warning occurs often then your model may be either severely ill-
conditioned or misspecified.

Chain 4

Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)

```
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.1 seconds.
```

```
All 4 chains finished successfully.
Mean chain execution time: 0.2 seconds.
Total execution time: 1.2 seconds.
```

```
# Check summary
fixed_fit$summary()
```

```
# A tibble: 16 x 10
  variable    mean  median    sd  mad      q5     q95  rhat ess_bulk ess_tail
  <chr>      <dbl>   <dbl> <dbl> <dbl>   <dbl>   <dbl> <dbl>   <dbl>   <dbl>
1 lp__      -232.   -232.   2.77  2.67  -237.   -228.   1.00   1349.   2032.
2 lambda[~    5.15    4.85  1.61  1.44    3.08    8.15   1.00   9674.   2754.
3 lambda[~    6.34    5.47  3.45  2.71    2.51   13.0    1.00  11346.   2978.
4 lambda[~    3.82    3.53  1.42  1.21    2.13    6.44   1.00   8911.   2892.
5 lambda[~    7.37    6.80  3.05  2.65    3.62   13.2    1.00   8951.   3177.
6 lambda[~    5.25    4.87  2.00  1.70    2.83    9.07   1.00   9917.   2932.
7 lambda[~    8.30    7.75  2.90  2.47    4.70   13.6    1.00   8763.   2850.
8 lambda[~    2.40    2.11  1.19  0.881   1.11    4.71   1.00   9665.   3015.
9 lambda[~    1.19    1.02  0.683  0.483   0.493    2.46   1.00   9835.   3117.
10 lambda[~    4.91    4.59  1.69  1.43    2.80    8.00   1.00   9132.   2774.
11 lambda[~    9.37    8.64  3.71  3.10    4.85   16.2    1.00   7799.   2925.
12 lambda[~    3.59    3.29  1.47  1.20    1.86    6.32   1.00   9786.   2780.
13 lambda[~    3.31    3.10  1.11  0.930    1.93    5.33   1.00   8572.   2778.
14 lambda[~    3.63    3.39  1.29  1.09    2.03    6.02   1.00  10759.   2703.
15 lambda[~    7.18    6.65  2.76  2.32    3.78   12.3    1.00   9633.   2629.
16 lambda[~    2.69    2.36  1.40  1.04    1.18    5.34   1.00   9038.   2880.
```

Checking MCMC for Q8

```
sum_fixed <- fixed_fit$summary()

sum_fixed[] <- lapply(sum_fixed, function(x) {
  if(is.numeric(x)) round(x, 3) else x
})

kableExtra::kable(sum_fixed)
```

variable	mean	median	sd	mad	q5	q95	rhat	ess_bulk	ess_tail
lp____	-	-	2.769	2.669	-	-	1.001	1348.748	2031.789
	232.168	231.813			237.185	228.231			
lambda[1]	5.145	4.845	1.614	1.440	3.082	8.150	1.000	9673.652	2753.764
lambda[2]	6.337	5.473	3.445	2.711	2.512	12.960	1.003	11346.057	2978.095
lambda[3]	3.821	3.527	1.418	1.209	2.131	6.443	1.000	8910.788	2892.072
lambda[4]	7.369	6.796	3.050	2.652	3.616	13.209	1.000	8951.409	3176.786
lambda[5]	5.247	4.872	2.000	1.697	2.827	9.067	1.002	9917.006	2931.685
lambda[6]	8.295	7.748	2.897	2.465	4.701	13.643	1.000	8762.661	2850.360
lambda[7]	2.396	2.112	1.193	0.881	1.106	4.713	1.000	9665.356	3015.185
lambda[8]	1.191	1.021	0.683	0.483	0.493	2.460	1.002	9835.107	3116.873
lambda[9]	4.913	4.591	1.692	1.429	2.797	8.001	1.000	9132.040	2774.387
lambda[10]	9.374	8.639	3.713	3.099	4.854	16.239	1.001	7799.122	2924.937
lambda[11]	3.594	3.289	1.473	1.203	1.861	6.316	1.002	9786.403	2780.094
lambda[12]	3.306	3.104	1.106	0.930	1.929	5.335	1.001	8572.162	2778.007
lambda[13]	3.630	3.387	1.289	1.085	2.033	6.019	1.002	10758.613	2702.731
lambda[14]	7.184	6.652	2.756	2.317	3.778	12.312	1.002	9633.121	2628.776
lambda[15]	2.689	2.356	1.399	1.043	1.179	5.343	1.001	9037.726	2879.664

The MCMC diagnostics suggest that the model converged well. All ($\hat{R}$) values are extremely close to 1, indicating good mixing and convergence across chains. In addition, both the bulk and tail effective sample sizes are large for all parameters, suggesting low autocorrelation and reliable estimation of posterior summaries and credible intervals. Therefore, there is no evidence of concerning convergence or sampling issues in the fitted model.

Q9

```
# Random effects sum
rand_sum <- fit$summary()

rand_cafe <- rand_sum |>
  filter(str_detect(variable, "^lambda\\[\"")) |>
  mutate(
    cafe = as.numeric(str_extract(variable, "\\d+")),
    model = "Random effects"
  ) |>
  select(cafe, model, median, q5, q95)

# Fixed effects sum
```

```

fixed_sum <- fixed_fit$summary()

fixed_cafe <- fixed_sum |>
  filter(str_detect(variable, "^lambda\\[") |>
    mutate(
      cafe = as.numeric(str_extract(variable, "\\d+")),
      model = "Fixed effects"
    ) |>
    select(cafe,model,median,q5,q95)

# Posterior median of overall mean expect waiting time
rand_draws <- as_draws_df(fit$draws(c("alpha", "beta")))

mu_draws <- rand_draws$alpha / rand_draws$beta
mu_median <- median(mu_draws)

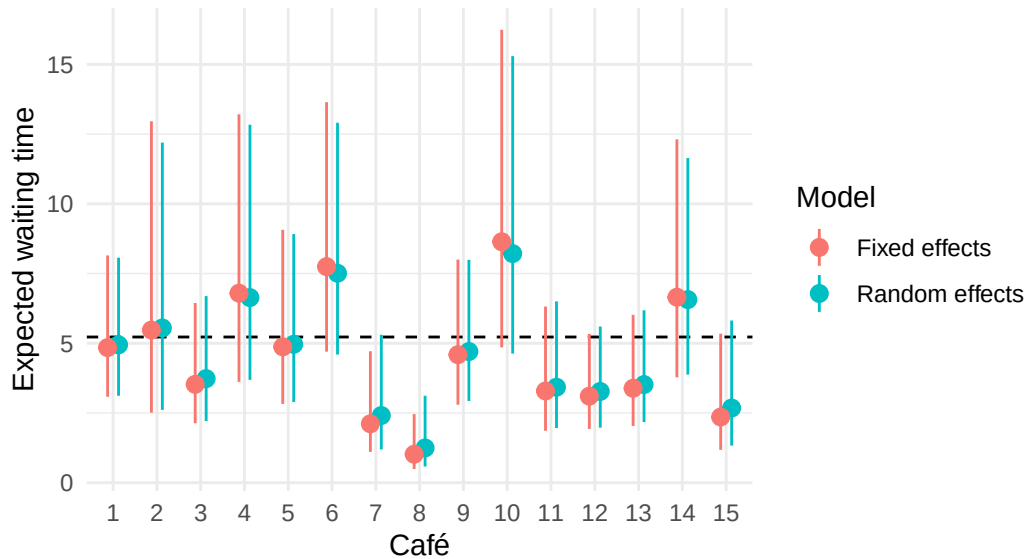
plot_data <- bind_rows(rand_cafe, fixed_cafe)

ggplot(plot_data, aes(x = factor(cafe), y = median, colour = model)) +
  geom_hline(
    yintercept = mu_median,
    linetype = "dashed",
    colour = "black"
  ) +
  geom_pointrange(
    aes(ymin = q5, ymax = q95),
    position = position_dodge(width = 0.5)
  ) +
  labs(
    x = "Café",
    y = "Expected waiting time",
    colour = "Model",
    title = "Comparison of fixed effects and random effects estimates",
    subtitle = "Posterior medians and 90% credible intervals"
  ) +
  theme_minimal()

```

Comparison of fixed effects and random effects estimates

Posterior medians and 90% credible intervals



The plot shows the general difference between the fixed effects and random effects models. The fixed effects model estimates each café separately, so its café-specific estimates are based mainly on the data from that café alone. As a result, the estimates can vary more strongly between cafés, especially for cafés with less data.

The random effects model partially pools information across cafés. This means that café-specific estimates are pulled towards the overall mean expected waiting time. Therefore, cafés with unusually high or low fixed effects estimates tend to have less extreme estimates under the random effects model. The random effects credible intervals may also be narrower or more stable because the model borrows strength across cafés.

The horizontal dashed line represents the posterior median of the overall mean expected waiting time from the random effects model. The random effects café estimates are generally shrunk towards this line, while the fixed effects estimates are allowed to move more freely away from it. This pattern is the main evidence of partial pooling in the random effects model.

Q10

In standard asymptotic theory, standard errors of parameter estimates typically shrink at a rate of $\frac{1}{\sqrt{n}}$ as the sample size increases. In this context, this means that the standard errors will initially decrease as we gain more information.

However, the mean waiting time parameter is designed to describe the whole population of cafés, and continuously sampling from only one café does not provide sufficient information

about the population as a whole. While repeated visits give increasingly precise information about Café Statistique itself, they provide little additional information about between-café variability. Therefore, the standard error for the population mean waiting time will decrease initially, but will not shrink to zero as n becomes very large, and will instead plateau.